

RAKIBUL ISLAM RAKIB

HARDWARE AND ELECTRONICS DEVELOPER

CONTACT

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TECHNICAL SKILLS

EMBEDDED SYSTEMS & ELECTRONICS

- ESP32 & Arduino development
- Custom PCB design and schematic layout
- Sensor interfacing
- Circuit design and hardware debugging

ROBOTICS & CONTROL SYSTEMS

- inverse kinematics implementation
- Closed-loop control
- PID control fundamentals
- Autonomous system logic development
- Drone flight controller configuration
- Power management for robotic systems
- In house PCB manufacturing

PROGRAMMING & DEVELOPMENT TOOLS

- C / C++ (Arduino framework)
- Python(Basic)
- Arduino IDE / PlatformIO
- Git & GitHub
- 3D Designing and printing (onshape)
- PCB Designing(Ki CAD , Easy EDA)

SOFT SKILLS

- Leadership & team management
- Technical training & mentoring
- Problem-solving mindset
- Event organization & coordination

ABOUT ME

Passionate about hardware and electronics developer with over 6 years of experience in robotics. Experienced in control systems, drones, autonomous robots, and custom hardware design, building complete systems from electronics to firmware and control interfaces. Looking forward to working with a supportive team to solve real-world problems and create practical solutions.

WORK EXPERIENCE

BRACU DUBURI (Electronics and Mechanical Team) 2025--Present

- Developed custom inhouse thrusters that has same capabalities as bluerobotics T200.
- Full electronics architecture design of DUBO MINI including mosfet based kill switch.
- Torpedo and dropper attachment design.
- Hull design and waterproofing up to 200m.

Tech Topia

Tech Envoy & R&D Team Member 2022-2024

- Started as a Tech Envoy, supporting and organizing technology workshops and programs through volunteer work.
- Contributed to the Research & Development (R&D) team, helping design and improve technical projects and learning materials.

Robodemy

Associate Trainer & R&D Developer 2024 - 2025

- Serving as an Associate Trainer, teaching robotics and electronics to students and mentoring them in hands-on projects.
- Member of the R&D team, developed advanced robotics products including two inverse-kinematics-based quadruped robots, including complete hardware architecture and firmware development.

DRMC Science Club

Vice President (Executive Panel 2025) 2024 - 2025

- Led the planning and execution of an international science and technology festival in April 2025.
- Managed event coordination, team organization, and program execution.

EDUCATION

Secondary School Certificate 2023

Annada Govt. High School | Cumilla Board

GPA: 5 .00

Higher Secondary Certificate 2025

Dhaka Residential Model College | Dhaka Board

GPA: 5 .00

Bachelor of Science (B.sc.) 2026 - Present

Electrical and Electronics Engineering | Brac University

Enrolled in Spring 2026

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LANGUAGES

- Bangla(Fluent)
- English(Fluent)

REFERENCE

AKM Badrul Hasan
Assistant Professor, Physics
Dhaka Residential Model College
Phone: 01922528717

ACHIEVEMENTS

- World Robot Olympiad 2025 **international round** winner(6th-full score)
- World robot olympiad 2025 open championship **national selection round** champion (gold medalist)
- World robot olympiad 2025 **regional round** champion
- World robot olympiad 2025 **national round** champion (gold medalist)
- World robot olympiad 2024 **international participant** -position 19th
- World robot olympiad 2024 **national round** champion (gold medalist)
- World robot olympiad 2023 **national round** winner(silver medalist)
- Robotronics fest 2025 runner up (**micromouse** segment)
- Robotronics fest 2025 runner up (**line following robot** segment)
- 15th Drmc national science carnival 2025 champion (**mini soccer** segment)
- 15th Drmc national science carnival 2025 champion (**fire fighter robot** segment)
- Init 4.0 runner-up (**line following robot** segment)
- 4th international robotech Olympiad 2024 champion (**line following robot** segment)
- 2nd in Green University EEE day(**line following robot** segment)
- 2nd in IEEE AUST Student Branch present OSCILLON (**line following robot** segment)
- 1st runner-up in UIU CSE Fest 2025(**line following robot** segment)
- 1st runner-up in EWU National RoboFest 2024(**line following robot** segment)
- 1st runner-up in Technomize 1.0 by KUET (**line following robot** segment)
- Drone race 1st runners up (DRMC science carnival 2026)
- Calibration 2.0 Champion(**micromouse** segment)
- Calibration 2.0 1st runners up(**micromouse** segment)
- Calibration 2.0 1st runners up(**line following robot** segment)
- Calibration 2.0 2nd runners up(**line following robot** segment)
- Winner codeavour Bangladesh 2024